Varun Iyer

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Education

University of Illinois Chicago — Aug 2022 – Present
PhD in Computer Science

Johns Hopkins University — Aug 2020 – May 2022
MSE in Computer Science

University of Massachusetts Amherst — Aug 2017 – May 2020
BS in Computer Science

Publications

* equal contribution

1. **Varun Iyer** and Cornelia Caragea. BLooP: Zero-Shot Abstractive Summarization Using Large Language Models with Bigram Lookahead Promotion. In *Proceedings of the Fifteenth Language Resources and Evaluation Conference*, 2026.

2. Kuan-Hao Huang, **Varun Iyer**, I-Hung Hsu, Anoop Kumar, Kai-Wei Chang, and Aram Galstyan. ParaAMR: A Large-Scale Syntactically Diverse Paraphrase Dataset by AMR Back-Translation. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2023.
(**Area Chair Award**)

3. Kuan-Hao Huang*, **Varun Iyer***, Anoop Kumar, Sriram Venkatapathy, Kai-Wei Chang, and Aram Galstyan. Unsupervised Syntactically Controlled Paraphrase Generation with Abstract Meaning Representations. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, 2022.

Experience

Research Assistant — Aug 2022 – Present
University of Illinois Chicago — *Chicago, IL*

- Working with **Professor Cornelia Caragea** on zero-shot abstractive document summarization
- Developing approaches to improve factuality in summarization
- Improved zero-shot ROUGE and BARTScore on CNN/DM, CCSum, Multi-News, and SciTLDR scientific abstracts

Teaching Assistant — Jan 2023 – Present
University of Illinois Chicago — *Chicago, IL*

- Led lab sections for courses in discrete mathematics and C programming
- Helped students understand NLP, discrete mathematics, and functional and systems programming

Applied Scientist Intern — May 2021 – Dec 2021
Amazon Alexa AI — *Remote*

- Worked with **Dr. Anoop Kumar** on unsupervised paraphrase-based data augmentation
- Used Abstract Meaning Representations to generate syntactically diverse paraphrases
- Achieved the best BLEU among the tested unsupervised syntactically controlled paraphrase generators on ParaNMT, Quora, PAN, and MRPC

Research Assistant — May 2020 – Dec 2021
Johns Hopkins University — *Remote*

- Worked with **Professor Benjamin Van Durme** on semantically grounded image classification
- Augmented a ResNet architecture with geometric hierarchical embeddings for few-shot learning
- Extended a neural entity-typing pipeline to new datasets and added distributed training

Undergraduate Researcher — Aug 2018 – May 2020
University of Massachusetts Amherst — *Amherst, MA*

- Worked with **Professor Andrew McCallum** on fine-grained entity typing using PyTorch
- Developed a stacked BiLSTM with embedding-based loss functions and hierarchical type constraints

Undergraduate Course Assistant Sep 2018 – May 2020
University of Massachusetts Amherst *Amherst, MA*

- Graded homework in Algorithms and Artificial Intelligence
- Helped students complete programming assignments written in Python and C

Undergraduate Research Intern May 2018 – Aug 2018
USC Information Sciences Institute *Marina del Rey, CA*

- Worked with **Professor Craig Knoblock** to build and link entities in a knowledge graph of space-related objects
- Implemented level-based access control for data across Elasticsearch indices
- Extracted information on thousands of satellites and incorporated it into the Elasticsearch workflow

Technical Skills

Languages: Python, Rust, C/C++, SQL
Frameworks: PyTorch, Transformers, vLLM, Kubernetes
Developer Tools: Git, Docker, GitLab CI/CD, tmux, VS Code, PyCharm, Helix
Libraries: Pandas, NumPy, Matplotlib, PyO3